

RUI SANG

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EDUCATION

Xi'an Jiaotong-Liverpool University

BSc (Hons) in Information and Computing Science

Suzhou, China

Sep 2022 – Jun 2026 (expected)

- Major GPA: 4.00 / 4.00 | Overall GPA: 3.96 / 4.00 | Top 5%
- Honors: University Academic Achievement Award Scholarship 2023, 2024, 2025
- Research Interest: Machine Learning, Signal Processing

PUBLICATIONS

1. R. Sang, Y. Liu, SceneGuard: Training-Time Voice Protection with Scene-Consistent Audible Background Noise, *AAAI 2025 Workshop*, 2025. [Under Review]
2. Y. Liu, R. Sang, P. Zhang, Z. Li, S. Li, Training a Perceptual Model for Evaluating Auditory Similarity in Music Adversarial Attack, *17th International Symposium on Computer Music Multidisciplinary Research (CMMR)*, 2025. [Accepted]
3. Y. Liu*, P. Zhang*, R. Sang, Z. Li, S. Li, MAIA: An Inpainting-Based Approach for Music Adversarial Attacks, *International Society for Music Information Retrieval (ISMIR)*, 2025. [Accepted]
4. P. Zhang*, Y. Liu*, Z. Li, R. Sang, Y. Cai, Y. Tan, S. Li, An Entropy-Guided Curriculum Learning Strategy for Data-Efficient Acoustic Scene Classification under Domain Shift, *Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE Workshop)*, 2025. [Accepted]

RESEARCH EXPERIENCE

Infrared-Visible Multimodal Image Fusion

Jun 2025 – Present

Research Assistant, Advised by Dr. Xiaohui Zhu

- Co-developed the SWF-DETR framework, architecting its cross-modal attention module to establish synergy between Transformer-based fusion and detection.
- Led experimental validation, improving mAP50 by 1.4% and model stability by 9.7% .

SceneGuard: Voice Cloning Defense via Scene-Consistent Noise

Jun 2025 – Sep 2025

Independent Research Project

- Proposed a training-time defense embedding scene-consistent audible noise into speech through a gradient-based masking strategy under SNR control.
- Achieved 5.5% speaker similarity degradation ($p < 10^{-15}$, Cohen's $d = 2.18$) while preserving 98.6% intelligibility (STOI = 0.986) and resisting 5 audio preprocessing attacks. [Pub. 1]

Entropy-Guided Curriculum under Device-Induced Shift

Jan 2025 – Jun 2025

Research Intern, Advised by Dr. Shencheng Li

- Trained an auxiliary domain classifier and applied Shannon entropy-based ranking to schedule data from invariant to device-specific.
- Achieved a +2.6% accuracy gain on unseen devices under a 5% low-data regime. [Pub. 4]

Reliability & Human-Aligned ML under Distribution Shift

Sep 2024 – Jun 2025

Research Intern, Advised by Dr. Shencheng Li

- Revealed perceptual misalignment in audio models through engineering 18,000+ perturbed samples, enabling a contrastive learning framework that improved robust accuracy by +9.15%. [Pub. 2]
- Engineered the critical importance analysis module for the MAIA white-box attack by adapting Grad-CAM to the audio domain, enabling targeted, perceptually-stealthy attacks. [Pub. 3]

PROJECTS

Online Meeting Room Booking System Team Lead

Mar 2025 – Apr 2025

- Led the full-stack architecture for a web application, defining the database schema, front-end UI, and a modular MVC backend with Spring Boot, JWT Security, and MySQL.
- Engineered a multi-level Redis caching strategy to optimize performance for hot data; containerized the application with Docker for consistent, high-availability deployment on Alibaba Cloud.

Secure File Transfer System Project Lead

Nov 2024 – Dec 2024

- Implemented client-side logic for the STEP protocol, managing socket communication, JSON serialization, and token-based authentication.
- Engineered a multi-threaded mechanism and implemented a thread-lock to guarantee data integrity.

SKILLS

Programming: Python, Java, C/C++, LaTeX.

Technologies: PyTorch, Spring Boot, MySQL, CUDA, Redis, Git, Docker, Figma.

ACTIVITIES

Class Committee Member: Class 4 of 2026, Xi'an Jiaotong-Liverpool University

2023 – 2024

Coordinated student-faculty communication, class activities, and student development initiatives.

Volunteer – Autism Support Program: "Star Children" Project, Suzhou

2023

Provided one-on-one support and facilitated inclusive community programs for children with autism.